

FOCUS.OS

Project Proposal for UCS503
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1 Higher-order goal

... secondary framing

This project aims to improve everyday productivity and task management for people who experience difficulties with planning, task initiation, prioritization, maintaining routines, and managing multiple responsibilities. The system is particularly intended to support users with ADHD and similar executive-function challenges, while remaining useful to a broader user population.

The higher-order goal is to reduce the cognitive effort required to decide what to do next and to help users make consistent progress on their responsibilities.

2 Time-to-value

... primary heuristic

- We will deliver a usable task-management and planning workflow within the first iteration.
- We will prioritize rapid user feedback through weekly iterations, allowing the planning and interaction model to evolve based on observed user behaviour and feedback.
- The initial system will focus on task capture, prioritization, and a simple “What should I do now?” workflow before introducing more advanced personalization and AI-based features.
- CI/CD will be introduced early to automate testing, building, and deployment of incremental versions.

3 Problem Statement

People often manage their responsibilities across multiple disconnected systems such as calendars, reminder applications, notes, habit trackers, and paper lists. Even when users know what they need to accomplish, deciding what to work on next and maintaining a realistic plan can become difficult.

This problem can be particularly significant for people experiencing executive-function difficulties, including difficulties commonly associated with ADHD.

Common pain points include:

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- **Task overload:** users may have many responsibilities but struggle to determine which task should be handled first.
- **Task initiation:** knowing that something needs to be done does not necessarily translate into starting it.
- **Large or vague tasks:** tasks such as “complete project” or “study for exam” can be difficult to start without smaller, actionable steps.
- **Rigid schedules:** missing one task can make a manually planned schedule unrealistic for the rest of the day.
- **Forgetting recurring responsibilities:** habits, routines, and repeated tasks may be difficult to maintain consistently.
- **Decision fatigue:** users may spend significant time deciding what to work on instead of actually working.

Why this matters: Existing productivity tools often provide task lists, calendars, reminders, or habit tracking, but the user may still be required to decide how to organize those tasks and what to do at a particular moment.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **web-based adaptive productivity and task-planning system** that allows users to put their responsibilities, routines, and recurring activities into one system. The system continuously organizes these inputs and helps the user determine the most appropriate next action.

The main components are:

- **Task management:** users can quickly add tasks along with deadlines, priority, and estimated duration.
- **Smart prioritization:** the system ranks tasks using factors such as urgency, priority, estimated duration, and available time.
- **Dynamic planning:** the system adapts the plan when tasks are delayed, skipped, completed early, or when the user’s available time changes.
- **Task breakdown:** larger tasks can be divided into smaller, actionable steps.
- **Routines and habits:** users can define recurring activities such as morning routines, exercise, reading, or other regular tasks.
- **Focus mode:** users can start a focused work session for the currently recommended task using a timer and progress tracking.
- **Gamification:** completion of tasks and focus sessions can provide points, levels, or user-defined rewards to encourage consistency.

4.2 Core workflow

1. User quickly captures the tasks and responsibilities they need to complete.
2. The system stores relevant information such as deadlines, priorities, estimated duration, and recurrence.
3. The system evaluates the user's available time and current tasks and determines which task should be addressed next.
4. The user starts the recommended task through Focus Mode.
5. After the task is completed, postponed, or interrupted, the system updates the remaining plan.
6. The system continuously adapts the schedule instead of requiring the user to manually rebuild the entire day's plan.

4.3 Example interaction

A user enters:

“Complete Software Engineering project by Friday.”

The system may break the task into smaller steps such as:

- Review project requirements
- Define functional requirements
- Design system architecture
- Design database
- Implement core functionality
- Test the system
- Prepare deployment and documentation

If the user later has only 30 minutes available, the system may recommend a small actionable step rather than presenting the entire project as one task.

5 Solution Approach

... Engineering focus

This is an engineering course project, so the focus will be on building a reliable adaptive planning workflow rather than developing a novel AI or clinical system.

5.1 Smart planning

The initial prioritization engine will use measurable factors such as:

- task deadline,
- user-defined priority,
- estimated duration,

- available time,
- task dependencies, where applicable,
- recurring commitments.

The system will provide an understandable reason for its recommendation, rather than making the prioritization decision completely opaque.

For example:

Work on DBMS assignment now

Due tomorrow, approximately 40 minutes remaining, and you currently have 50 minutes available.

5.2 Adaptive scheduling

The system will maintain a dynamic plan rather than treating the schedule as fixed. When a task is missed or interrupted, remaining tasks can be reordered based on their current urgency, priority, and available time.

5.3 AI-assisted functionality

AI may be used as a supporting component for tasks such as:

- converting vague tasks into actionable subtasks,
- suggesting task duration,
- assisting with schedule creation,
- suggesting an appropriate next action,
- adapting plans based on user-provided context.

AI functionality will be introduced only where it provides measurable value and will not be treated as the primary novelty of the project.

5.4 Web architecture

A typical 3-tier architecture will be used:

- **Frontend:** task dashboard, planner, routines, focus mode, and progress views.
- **Backend API:** authentication, task management, prioritization, scheduling, habit/routine logic, and progress tracking.
- **Database:** users, tasks, deadlines, routines, habits, focus sessions, task history, and reward information.

5.5 CI/CD and fast delivery enabler

CI/CD will be used to:

- automatically build and test the application,
- run unit and integration tests on changes,
- produce repeatable deployment artifacts,
- enable frequent deployment of incremental features.

6 Evaluation Criterion

... measurable and attributable

The effectiveness of the system will be evaluated using measurable user-facing and system-level metrics.

6.1 Primary evaluation metrics

- **Task initiation time:** time between a recommended task being presented and the user starting the task.
- **Planning effort:** time required for users to decide what task to work on compared with a manual planning approach.
- **Task completion rate:** percentage of planned tasks completed within the intended time period.

6.2 Secondary evaluation metrics

- **Schedule recovery:** percentage of delayed schedules that successfully generate a revised plan without manual replanning.
- **Recommendation usefulness:** user rating of whether the recommended next task was appropriate.
- **Task breakdown usefulness:** user rating of whether generated subtasks made large tasks easier to start.
- **Focus-session completion:** percentage of started focus sessions completed successfully.
- **User satisfaction:** simple 1–5 rating collected during pilot testing.

6.3 Pilot validation plan

... high-level

- Recruit a small group of users for initial usability testing.
- Compare manual planning with the proposed system for representative task-management scenarios.
- Collect task initiation time, completion rate, planning effort, and user feedback.
- Use the results to refine the prioritization and interaction workflow in subsequent iterations.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical contexts.

1. **Scalable (theoretically):** The system should support an increasing number of users and tasks through:
 - stateless backend APIs,
 - indexed database queries,
 - pagination for large task histories,
 - separation of frontend and backend services.
2. **Operationally deployable (practically):** The application should be deployable using common web hosting, managed databases, and automated CI/CD pipelines.

8 Engine availability heuristic

A mature set of engineering tools and services is available for implementing the proposed system:

- Web framework for the frontend and backend API.
- Relational or document database for persistent user data.
- Authentication and authorization mechanisms.
- Scheduling and notification services.
- AI/LLM APIs for optional task decomposition and planning assistance.
- Test frameworks for unit and integration testing.
- CI runner and deployment platform.

These mature components allow the project to focus on software engineering, workflow design, system reliability, user feedback, and measurable outcomes rather than research into new AI algorithms.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- User registration and basic profile
- Quick task creation
- Task deadlines, priorities, and estimated duration
- Basic task prioritization
- “What should I do now?” recommendation
- Focus timer
- Basic task completion tracking
- CI: build + backend unit tests + linting
- CD to staging

9.2 Subsequent deliverables

- Dynamic schedule adjustment after missed or interrupted tasks
- Task decomposition into smaller actionable steps
- Recurring routines and habits
- AI-assisted planning and task breakdown
- Gamification and user-defined rewards
- Improved personalization using historical task behaviour
- Notification and quick-access mechanisms
- Calendar integration, if feasible within the project timeline

10 Risks and mitigations

- **App avoidance:** provide quick task capture, minimal interaction, and optional notifications/widgets so users do not need to repeatedly navigate through the application.
- **Information overload:** keep the primary interface focused on the single most useful question: “What should I do now?”
- **Incorrect prioritization:** provide transparent reasons for recommendations and allow users to override them.
- **Inaccurate time estimates:** allow users to modify estimates and use historical completion data for future suggestions.
- **Repeated procrastination:** identify frequently postponed tasks and offer smaller actionable steps or rescheduling.
- **Schedule disruption:** dynamically recalculate remaining tasks when the user falls behind or available time changes.
- **Notification fatigue:** prioritize important notifications rather than generating reminders for every activity.
- **Feature overload:** develop the core task-planning workflow first and treat habits, gamification, and advanced AI functionality as subsequent iterations.
- **Privacy:** minimize collection of personal information and provide appropriate controls for task, routine, and behavioural data.
- **ADHD-related claims:** position the system as a productivity and planning tool designed with executive-function challenges in mind, rather than as a medical or therapeutic product.

11 Summary

This project proposes a web-based adaptive task-planning and focus-support system designed around the difficulties of managing multiple responsibilities, initiating tasks, prioritizing work, and recovering from disrupted schedules.

Rather than functioning only as a conventional task list, the system will continuously organize user inputs and provide a context-aware recommendation for what the user should work on next. The system will progressively incorporate task breakdown, routines, focus sessions, adaptive scheduling, AI assistance, and gamification.

The project meets the course criteria by emphasizing rapid incremental delivery, CI/CD-supported development, mature engineering components, scalability, and measurable evaluation through task initiation time, completion rate, planning effort, and schedule recovery.